Management information system (MIS)

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- Chapter 2: FOUNDATIONAL CONCEPTS IN MIS
- Chapter 3:Information System Development and system life cycles
- Chapter 4: COMMON BUSINESS APPLICATIONS OF INFORMATION TECHNOLOGY
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Chapter 1
Introduction

Definition of MIS

Introduction

- Every aspect of management in the modern age relies heavily on information to thrive.
- It is generally believed that information is power and those who have it have a power.

- In the past three or four decades, a fast changing circumstances and environments have necessitated the need for the proper dissemination of appropriate information at various levels of management so that:
  - It lead to better planning, better decision making and better

Introduction Cont.

- Today the need for updated information has become inevitable to arrive at an effective decision in all walks of life.
- Whether it is industry, commerce, defense, banking, education, economics or politics, information is needed everywhere.
- Information technologies including Internet-based information systems, are playing a vital and expanding role in business.
- Improve the efficiency and effectiveness of:
  - Business processes
  - Managerial decision making
  - Workgroup collaboration
  - Increase their competitive positions in a rapidly

Basic Concepts

Data

- Data is the most basic element of any information system.
- It is raw facts – recorded, unformatted information, such as words and numbers.
- Has no meaning in and of itself.
- E.g. Elementary description of things, events, activities and transactions that are recorded, and stored but are not organized to convey any specific meaning.

Basic Concepts Cont.

Information

- Is a representation of reality, which is processed data. It is also data that have been shaped into a form that is meaningful and useful to human beings.
- The data which is organized and presented at a time and place so that the decision-maker may take necessary act.
- It consists of data that has been retrieved, processed or otherwise used, for informative or inferential purposes, arguments or as a basis for forecasting.

Basic Concepts Cont.

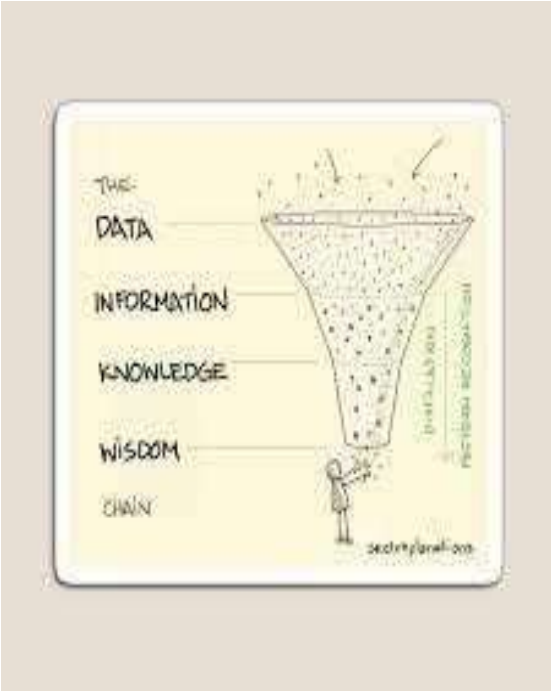
Knowledge

- Is defined as what is learning, understood and aware. (factual, conceptual and procedural knowledge).
- The fact of knowing, understanding and familiarity with subject, procedures, place, etc.

Wisdom

- Is the ability to know what is true or right, common sense or the collection of knowledge. (Emotion regulation, moral reasoning, compassion, humanity, patience, dealing with uncertainty.
- Is accumulated knowledge that goes beyond knowledge by representing broader, more generalized rules and schemas for understanding a specific domains.

Basic Concepts Cont.



Definition of MIS

- MIS is a system or process that provides the information necessary to manage an organization effectively.

Management: is the art of getting things done through and with the people in formally organized groups.

Information :is data that is processed and is presented in a form which assists decision maker

System: is defined as a set of elements which are joined together to achieve a common objective

MIS should have a clearly defined framework of guidelines, policies or practices, standards and procedures for the organization.

- MIS is A system that collects, process, stores the data and distributes information to help in decision making for managerial function.
- Right Information-To the right person -At the right place- At the right time- In the right Form- At the right cost

MIS Model

- A model is an abstraction of something it represents (some phenomenon), called an entity. For example, if a model represents a firm, then the firm is an entity.

Types of Models:

1. Physical Models:

- It is a model that exists in three-dimensional form. For example, clay model of a new automobile developed by designers.

2. Narrative Models:

- It is created by verbal or written description. Also, these can be created anywhere without much resources hence they are most popular.

3. Graphs:

- It is usually a two-dimensional diagram. For example, graphs and charts.

MIS Model Cont.

4. Mathematical Models:

- It uses symbolic notations and mathematical equations to represent a system. It can be represented by 3-D also. Following are the types of mathematical models:

(a)Influence of Time: Static Model does not include time as a variable whereas, Dynamic Model allows the changes of system attributes to be derived as a function of time.

(b)Degree of Certainty: A probability is a chance of occurring something will happen. Probabilities ranges from 0 to 1. A model that includes probabilities is called Probabilistic model, otherwise it is called a Deterministic model.

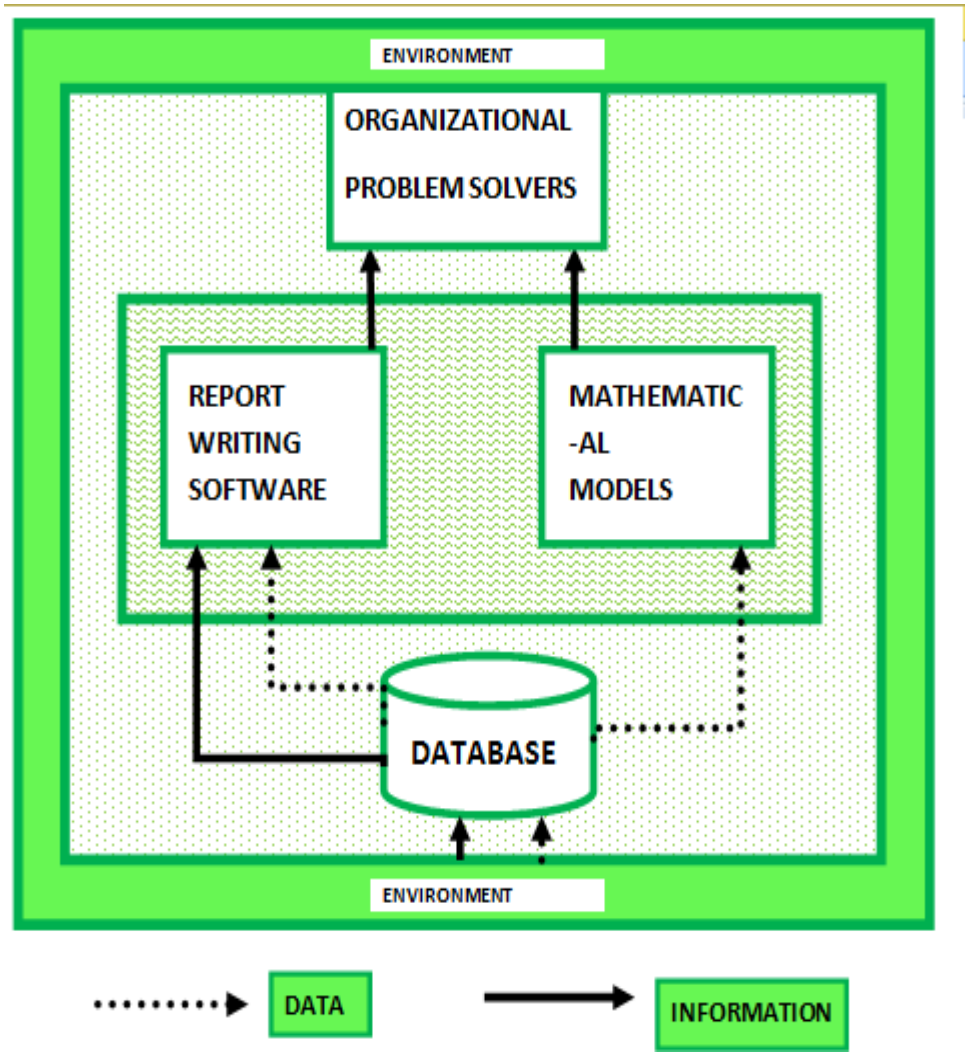
(c)Ability to Achieve Optimization: Optimizing Model(transportation, assignment, LPP etc) selects the best solution among the alternatives.

MIS Model Cont.

Steps of Model Construction:

i. Define the problem and the factor that influence it.
ii. Select criteria to guide the decision and establish objectives.
iii. Formulate a model that helps management to understand the relationships between the influence factors and the objectives the firm is trying to achieve.
iv. Collect the relevant data while trying to avoid the incorporation of superfluous information into the model.
v. Identify and evaluate the alternatives.
vi. Select the best alternative.
vii. Implement the alternative.

MIS Model Cont.



Components of MIS Model:

- **Organizational Problem Solvers**: outputs are used by persons who are responsible for solving the firm's problems(say managers).
- **Database**: it's contents are used by software that produces report as well as mathematical model.
- **Mathematical Model**: it produces information as a simulation of firm's operation.
- **Report Writing Software**: It produces
  both periodic and special report.
- **Environment**: is of great importance to the firm. It is the very reason of the firm's existence.

Components of Information System

- The basic components of information systems are listed below. Note that not every system includes all these components.

1.Resources of **people**: (end users and IS specialists, system analyst, programmers, data administrators etc.).

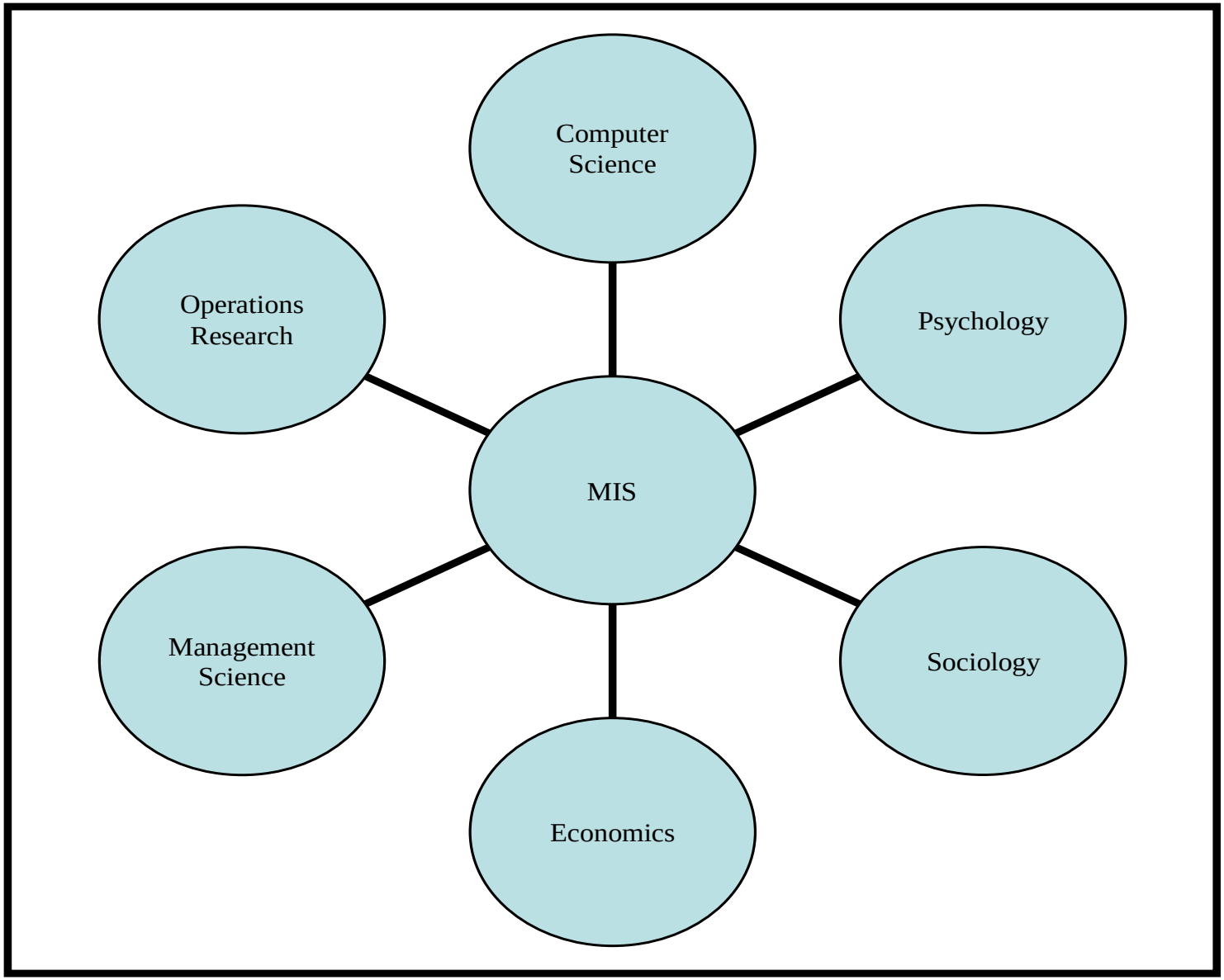
2.Hardware: (Physical computer equipment's and associate device, machines and media).

3. Software: (programs and procedures).

4. Data: (data and knowledge bases), and

5. Networks: (communications media and network support).

Contemporary Approaches To MIS



The Characteristics of the Information Age

- The information age is distinguished from previous ages on the basis of five characteristics.
- The Information Age came about with the rise of an information based society.
- Businesses in the Information Age depend on information technology to get their work done.
- In the Information Age, work processes are transformed to increase productivity.
- Success in the Information Age is largely determined by the effectiveness with which information technology is used.
- In the Information Age, information technology is embedded in many products and services.

CONTD.

- An Information Society: The Information Age came about with the rise of an information society. In an information society, more people work at handling information than at agriculture and manufacturing combined. This is true in most advanced western countries.
- ***Dependence on Information Technology:*** In the information age, businesses depend on information technology. Furthermore, an information society depends on much more than computers alone. Knowing when to use computers is as important as knowing how to use them. Equally important are the abilities to communicate information using computers and to interconnect people through information technology.
- ***Transformation of Work:*** In addition to providing new tools, the Information Age transforms earlier tools and work processes by increasing their productivity and effectiveness. Work processes are the activities that workers perform, the way they perform these activities, and the tools they use. *Productivity* is a measure of the amount of work that can be accomplished with a given level of effort. *Effectiveness* is the extent to which desirable results are achieved.

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- ***Information Technology Influences Successes:*** In the Information Age**,** the most successful are those who know how to make the most of information technology. Using information technology means much more than just knowing how to key data into a computer or how to print reports. It also means knowing what IT can improve your personal performance and how it can exchange a business's products and services in a way that increases their value to the firm's customers.
- ***Embedded Information Technology:*** In the Information Age, information technology is embedded in many products and services. Keep in mind, however, that knowledge-based products are desirable only if the knowledge adds extra value to the product. Value may be convenience, quality, reliability, or novelty--- any characteristic the consumer feels is useful. Currently, many products are affected by the information technology such as travel, automobile, television, aircraft, etc.

Thank you